## 03-SENSOR-MODULES

| Item # | Sensor Module Name | Technical Specifications & Model Details | Required Quantity | Asset Classification |
|---|---|---|---|---|
| 01 | **Digital Temperature & Humidity (DHT11)** | Single-bus digital output sensor module with pull-up resistor onboard; temp range 0–50°C; humidity range 20–90% RH; supply 3.3V–5V | **10 Units** | **Consumable** |
| 02 | **Analog Humidity Sensor Module** | Resistance-based relative humidity sensor probe board; range 20–95% RH; linear analog voltage response | **10 Units** | **Consumable** |
| 03 | **Soil Moisture Sensor Module** | Double-sided nickel-plated fork probe with LM393 comparator breakout module; analog and digital outputs | **8 Units** | **Consumable** |
| 04 | **Raindrop Detection Board** | 50×40 mm nickel-plated rain sensor board with separate control module (LM393 comparator and trimmer) | **8 Units** | **Consumable** |
| 05 | **Turbine Water Flow Sensor** | Hall-effect liquid flow meter; 1–30 L/min range; 5V DC; pulse frequency output (f = 7.5× Q) | **5 Units** | **Consumable** |
| 06 | **PIR Human Motion Detector (HC-SR501)** | Pyroelectric infrared sensor module; dome Fresnel lens; range 3–7m; adjustable delay and sensitivity; 3.3V logic output | **6 Units** | **Consumable** |
| 07 | **Ultrasonic Distance Sensor (HC-SR04)** | 40 kHz ultrasonic transmitter and receiver module; range 2 cm – 400 cm; precision 3mm; 4-pin header (VCC, Trig, Echo, GND) | **12 Units** | **Consumable** |
| 08 | **IR Proximity / Obstacle Avoidance Module** | 940nm IR emitter and photodiode pair; onboard LM393 comparator; detection distance 2–30 cm; 35° angle | **8 Units** | **Consumable** |
| 09 | **8-Channel IR Tracking Sensor Array** | 8-sensor reflective infrared bar with onboard comparators; 8 individual digital output pins for line-following robotics | **10 Units** | **Consumable** |
| 10 | **Sound Detection Microphone Module** | Electret condenser microphone with LM393 amplifier and digital threshold sensitivity trimmer | **5 Units** | **Consumable** |
| 11 | **High-Sensitivity Big Sound Module** | Large electret microphone capsule module with dual analog sound waveform and digital trigger outputs | **5 Units** | **Consumable** |
| 12 | **Capacitive Touch Module (TTP223B)** | Single-key capacitive digital touch switch; momentary / toggle output mode selectable; supply 2.0V–5.5V DC | **10 Units** | **Consumable** |
| 13 | **Touch Sensor Module (TTP22X)** | Multi-purpose capacitive contact sensor with low standby power; operating voltage 2.4V–5.5V DC | **6 Units** | **Consumable** |
| 14 | **Metal Touch Sensor Module (KY-036)** | Metal touch sensor with Darlington transistor and comparator; senses human skin touch on metal lead | **6 Units** | **Consumable** |
| 15 | **12-Channel Capacitive Touch (MPR121)** | MPR121 capacitive touch controller; 12 independent sensing inputs; I2C digital interface; 3.3V logic | **5 Units** | **Consumable** |
| 16 | **Pulse Heart-Rate Sensor** | Optical photoplethysmography (PPG) pulse sensor with built-in ambient noise cancellation; 3.3V–5V DC | **5 Units** | **Consumable** |
| 17 | **Color Recognition Sensor (TCS3200)** | TCS3200 programmable color light-to-frequency converter with 4 white LEDs; reads Red, Green, Blue, Clear filters | **5 Units** | **Consumable** |
| 18 | **ADXL345 3-Axis Digital Accelerometer Module** | 3-axis digital accelerometer; high resolution (13-bit) measurement up to ± 16g; ultra-low power; SPI and I2C digital interfaces; 3.3V/5V compatible | **5 Units** | **Consumable** |

| Item # | Sensor Module Name | Technical Specifications & Model Details | Required Quantity | Asset Classification |
|---|---|---|---|---|
| 19 | **MPU6050 6-Axis Gyroscope and Accelerometer Sensor Module with I2C Interface and Digital Processor, for Robotics, Metal and Plastic** | 6-Axis motion tracking module integrating 3-axis gyroscope (± 250, ± 500, ± 1000, ± 2000^°/s) and 3-axis accelerometer (± 2g, ± 4g, ± 8g, ± 16g) with onboard Digital Motion Processor (DMP); I2C interface; operating voltage 3.3V–5V DC with onboard low-dropout regulator; essential for self-balancing robots, gesture navigation, flight stabilization, metal and plastic chassis robotics | **10 Units** | **Consumable** |
| 20 | **3-Axis Digital Magnetometer** | 3-axis magnetoresistive compass module (HMC5883L / QMC5883L); I2C interface; 12-bit ADC; ± 8 Gauss range | **10 Units** | **Consumable** |
| 21 | **Laser Diode Emitter Module** | 650nm red point laser diode module with onboard current-limiting resistor; output 5mW; 5V DC | **10 Units** | **Consumable** |
| 22 | **MQ-2 Smoke & Combustible Gas Sensor** | Wide-spectrum gas sensor module for methane, LPG, smoke, hydrogen; 5V heater with analog/digital comparator outputs | **3 Units** | **Consumable** |
| 23 | **MQ-3 Alcohol Vapor Gas Sensor** | High sensitivity to ethanol alcohol vapor; ideal for breathalyzer circuits; onboard LM393 comparator | **3 Units** | **Consumable** |
| 24 | **MQ-4 Methane & Natural Gas Sensor** | High sensitivity to methane (CH_4) and natural gas leaks; fast response; onboard LM393 comparator | **3 Units** | **Consumable** |
| 25 | **MQ-5 LPG & Town Gas Sensor** | High sensitivity to LPG and compressed natural gas (CNG); onboard LM393 comparator | **3 Units** | **Consumable** |
| 26 | **MQ-6 Propane & Butane Gas Sensor** | Specialized high sensitivity for domestic LPG and butane gas cylinder leak detection; onboard LM393 comparator | **3 Units** | **Consumable** |
| 27 | **MQ-7 Carbon Monoxide (CO) Sensor** | High sensitivity to toxic carbon monoxide gas; dual-temperature cycling heater; onboard LM393 comparator | **3 Units** | **Consumable** |
| 28 | **MQ-8 Hydrogen Gas Sensor** | Specialized sensitivity to pure hydrogen gas (H_2); low sensitivity to alcohol/smoke; onboard LM393 comparator | **3 Units** | **Consumable** |
| 29 | **MQ-135 Air Quality & Pollution Sensor** | Environmental air quality detector for ammonia (NH_3), NO_x, alcohol, benzene, and smoke; onboard LM393 | **3 Units** | **Consumable** |